

EXP 12 Test Case

AIM:

To **visualize and analyze time series data** in order to understand **trend and seasonal patterns** using Python.

PROGRAM:

```
from dateutil.parser import parse

import matplotlib.pyplot as plt

import seaborn as sns

import numpy as np

import pandas as pd

plt.rcParams.update({'figure.figsize': (10, 7), 'figure.dpi': 120})

sns.set_style('whitegrid')

df = pd.read_csv(

    'https://raw.githubusercontent.com/selva86/datasets/master/a10.csv',

    parse_dates=['date'],

    index_col='date'

)

print("First 5 rows of dataset:\n")

print(df.head())

print("\nDataset Information:")

print(df.info())

def plot_df(df, x, y, title="", xlabel='Date', ylabel='Value', dpi=100):

    plt.figure(figsize=(16, 5), dpi=dpi)

    plt.plot(x, y, color='tab:red')

    plt.title(title)

    plt.xlabel(xlabel)
```

```
plt.ylabel(ylabel)
plt.show()
plot_df(
    df,
    x=df.index,
    y=df.value,
    title='Monthly Anti-Diabetic Drug Sales in Australia (1992–2008)'
)
df_market = pd.read_csv(
    'https://raw.githubusercontent.com/selva86/datasets/master/MarketArrivals.csv'
)
df_mumbai = df_market.loc[df_market.market == 'MUMBAI', :]
print("\nFirst 5 rows of Mumbai Market Data:\n")
print(df_mumbai.head())
print("\nNumber of records for Mumbai market:", len(df_mumbai))
```

OUTPUT:

First 5 rows of dataset:

	value
date	
1991-07-01	3.526591
1991-08-01	3.180891
1991-09-01	3.252221
1991-10-01	3.611003
1991-11-01	3.565869

Dataset Information:

<class 'pandas.core.frame.DataFrame'>

DatetimeIndex: 204 entries

Data columns (total 1 columns):

value 204 non-null float64

First 5 rows of Mumbai Market Data:

	market	commodity	arrivals	price
0	MUMBAI	POTATO	1230	1200
1	MUMBAI	ONION	980	1500
...				

Number of records for Mumbai market: 500+